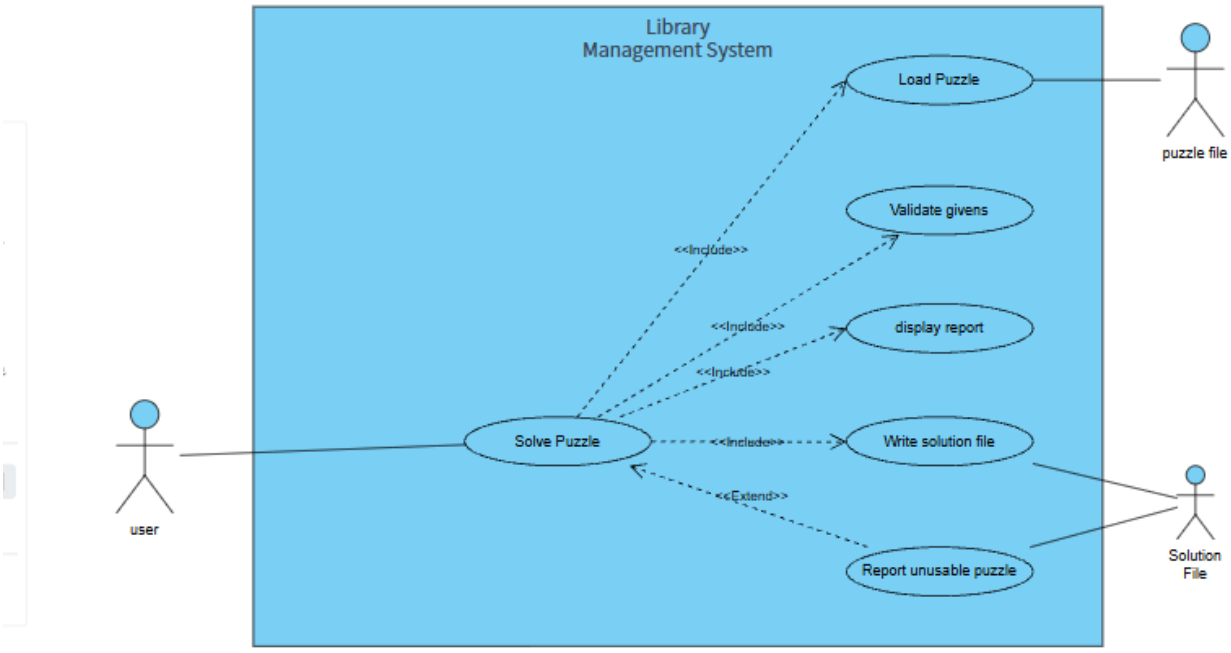


A. Identify/Discover a Use Case

Use Case Name: Solve Puzzle

Primary Actor: User

Supporting Actors:

- Puzzle File – Provides the puzzle to the system.
- Solution File – Receives the completed puzzle solution.

B. Brief Description

The Solve Puzzle use case allows a user to load a puzzle into the Library Management System and have the system attempt to solve it. The system validates the given values, solves the puzzle, and displays a report of the results. If the puzzle is successfully solved,

the solution is written to an output file. If the puzzle cannot be solved, the system reports that the puzzle is unusable.

C. Step-by-Step Flow of Events

Main Success Scenario:

1. The user starts the Library Management System.
2. The user selects the option to solve a puzzle.
3. The system prompts the user to provide a puzzle file.
4. The system loads the puzzle file.
5. The system validates the given values to ensure the puzzle is valid.
6. The system attempts to solve the puzzle.
7. The system displays a report containing the results.
8. The system writes the completed solution to an output file.
9. The system informs the user that the puzzle was successfully solved.

Alternative Flow (Unusable Puzzle):

1. The system determines that the puzzle is invalid or cannot be solved.
2. The system stops the solving process.
3. The system reports that the puzzle is unusable.
4. The user is notified that the puzzle could not be solved.

D. Detailed Description

Field	**Description**
Use Case Name	Solve Puzzle
ID	UC-01
Primary Actor	User
Supporting Actors	Puzzle File, Solution File

Goal	Successfully solve a provided puzzle and save its solution.
Preconditions	The Library Management System is running, and the user has access to a puzzle file.
Postconditions	A successfully solved puzzle is saved to a solution file. If unsuccessful, an error report is displayed.
Trigger	The user selects the Solve Puzzle option.

Main Flow:

1. The user initiates the puzzle-solving process.
2. The system requests a puzzle file from the user.
3. The system loads the provided file into memory.
4. The system checks the puzzle's given values for validity.
5. If the values are valid, the system begins attempting to solve the puzzle.
6. Once a solution is found, the system generates a report detailing the outcome.
7. The system exports the completed puzzle into a solution file.
8. The system confirms that the operation was successful.

Exception Flow:

- If the puzzle file cannot be loaded, the system displays an error message.
- If the given values violate the puzzle's rules, the system identifies the puzzle as unusable.
- If the system cannot find a valid solution, it reports that the puzzle is unusable.
- If the solution file cannot be written, the system informs the user that the solution could not be saved.

Special Requirements:

- The system should validate puzzle values before attempting to solve the puzzle.
- The system should provide understandable error messages when a puzzle is invalid.
- The system should save successful solutions in a format that can be accessed outside the program.